Assignment 2

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Problem 1: Learning word embeddings from IIT Jodhpur data.

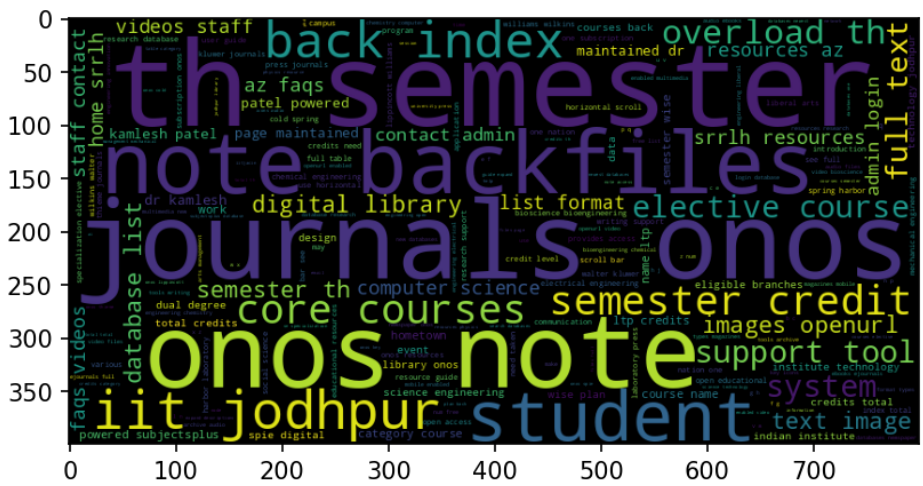
➔ Dataset preparation

First, I did web crawling using "**requests beautifulsoup4 lxml**". I get large Data from the "iitj.ac.in" website, which contains metadata with webpage content (like: URL, title of webpage, wordcount per page). And then I did preprocessing on the data to remove metadata, stopwords, punctuation and convert into lower case. Also, remove other language content, garbage or meaningless words. So it becomes small and useful for the task.

Data Statistics:

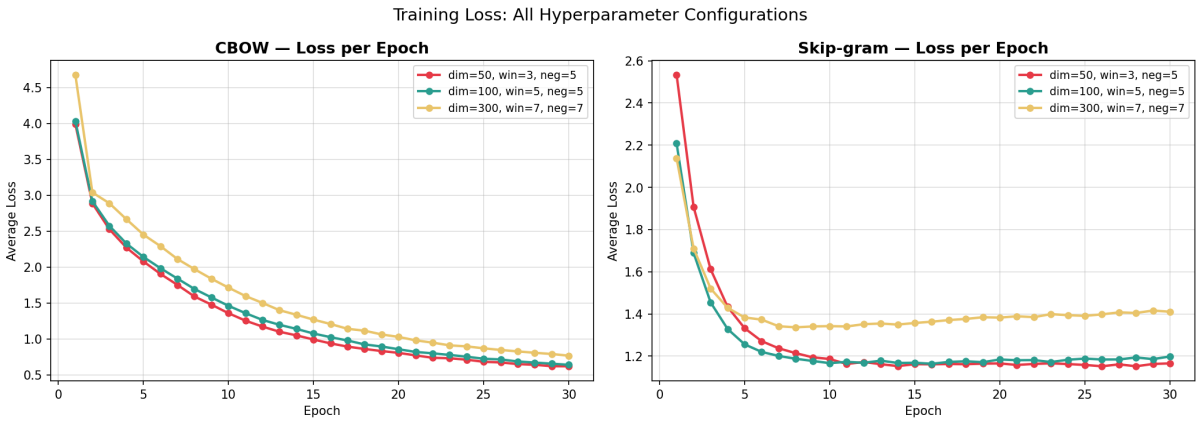
- **Total tokens: 54,204**
- **Unique tokens: 7,524**
- **Sample words:**
  ['student', 'wellbeing', 'committee', 'iit', 'jodhpur', 'embark', 'journey', 'aasmaan', 'limitless', 'sky', 'wellbeing', 'newsletter', 'previous', 'next', 'mental', 'health', 'state', 'mental', 'wellbeing', 'enables', 'people', 'cope', 'stresses', 'life', 'realise', 'abilities', 'learn', 'well', 'work', 'well', 'contribute', 'community', 'mental', 'health', 'basic', 'human', 'right', 'crucial', 'personal', 'community', 'socioeconomic', 'development', 'world', 'health', 'organization', 'important', 'physical', 'health', 'connected', 'body']
- **Vocabulary size: 2,055 words**
- Top 10 words : ['onos', 'engineering', 'note', 'semester', 'journals', 'research', 'access', 'library', 'th', 'jodhpur']

Wordcloud



➔ Model Training

Loss curve:



➔ I did experiment on this configuration:

```
CONFIGS = [
    ('Config-A (dim=50,  win=3, neg=5)',  50,  3,  5),
    ('Config-B (dim=100, win=5, neg=5)', 100,  5,  5),
    ('Config-C (dim=300, win=7, neg=15)',300,  7, 7),
]

EPOCHS          = 30
LEARNING_RATE   = 0.1
MIN_WORD_COUNT  = 5     # ignore words below this frequency
SUBSAMPLING_T   = 1e-3
```

➔ Analysis

Architecture	Config	Dim	Win	Neg	Loss	Time (s)
CBOW	Config-A (dim=50, win=3, neg=5)	50	3	5	0.6205	31.90
Skip-gram	Config-A (dim=50, win=3, neg=5)	50	3	5	1.1657	88.18
CBOW	Config-B (dim=100, win=5, neg=5)	100	5	5	0.6410	34.71
Skip-gram	Config-B (dim=100, win=5, neg=5)	100	5	5	1.1986	130.77
CBOW	Config-C (dim=300, win=7, neg=15)	300	7	15	0.7679	41.49
Skip-gram	Config-C (dim=300, win=7, neg=15)	300	7	15	1.4105	199.54

(where **Dim**: Embedding dimension, **Win**: Context window size, **Neg**: Number of negative samples)

➔ The top 5 nearest neighbours for the words:

research	1	governments	0.6251
	2	outstanding	0.5816
	3	notforprofit	0.5778
	4	corequisite	0.5778
	5	paper	0.5522
student	1	every	0.6121
	2	heads	0.5749
	3	experiences	0.5672
	4	needs	0.5621
	5	fests	0.5561
phd	1	amit	0.6499
	2	prof	0.6418
	3	graduate	0.6086
	4	bombay	0.5918
	5	postdoctoral	0.5908
exam	1	admission	0.7081
	2	msc	0.6789
	3	select	0.5974
	4	preparatory	0.5773
	5	edition	0.5752
university	1	cambridge	0.7202
	2	oxford	0.6912
	3	allahabad	0.6553
	4	press	0.6485
	5	bengaluru	0.6103

CBOW

research	1	ghosh	0.5682
	2	fellow	0.5290
	3	grant	0.5195
	4	work	0.5186
	5	minister	0.5159
student	1	ltp	0.6788
	2	heads	0.6002
	3	category	0.5837
	4	overload	0.5823
	5	whereas	0.5764
phd	1	postdoctoral	0.6966
	2	amit	0.6817
	3	usa	0.6356
	4	fellow	0.6172
	5	institutes	0.6008
exam	1	admission	0.6945
	2	master	0.6086
	3	undergraduate	0.5986
	4	choose	0.5894
	5	preparatory	0.5733
university	1	cambridge	0.6737
	2	jfm	0.6426
	3	press	0.6218
	4	nb	0.6018
	5	ecsu	0.5991

Skip-gram

➔ Semantic Meaningfulness Analysis

```
================================================================
  ANALOGY EXPERIMENTS — CBOW
  Formula: a − b + c = ?
================================================================
  Analogy                          Rank  Prediction          Score  Meaningful?
  ----------------------------------------------------------------
  UG:BTech :: PG:?                  1    mba                 0.6017  ✓ yes
                                    2    dcs                 0.5976
                                    3    awarded             0.5871

  student:exam :: professor:?       1    msc                 0.5524  ✓ yes
                                    2    solidworks          0.5443
                                    3    junior              0.5330

  PhD:thesis :: MTech:?             1    curriculum          0.5917  ✓ yes
                                    2    electives           0.5748
                                    3    mo                  0.5499

  research:lab :: student:?         1    team                0.4907  ~ maybe
                                    2    required            0.4595
                                    3    fests               0.4450
```

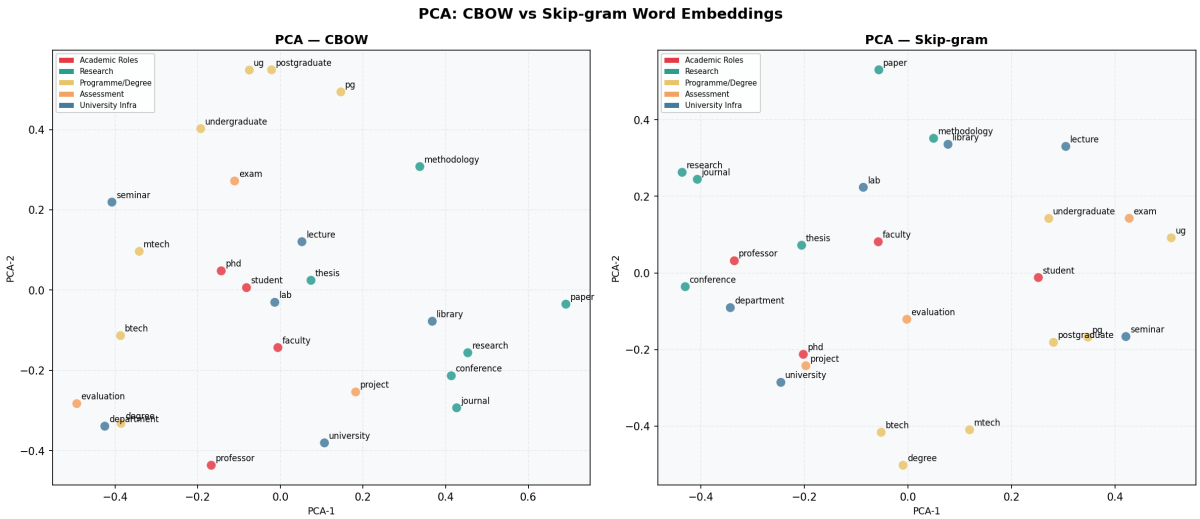
- **CBOW:**
  Produces more **contextually relevant outputs** (e.g., UG → MBA, PhD → curriculum), showing better understanding of academic relationships. However, some predictions are noisy (dcs, mo).

```
==================================================================
  ANALOGY EXPERIMENTS — Skip-gram
  Formula: a − b + c = ?
==================================================================
  Analogy                          Rank  Prediction          Score  Meaningful?
  ------------------------------------------------------------------
  UG:BTech :: PG:?                 1     mtech              0.4892  ~ maybe
                                   2     program            0.4879
                                   3     technical          0.4696

  student:exam :: professor:?      1     master             0.5860  ✓ yes
                                   2     msc                0.5554
                                   3     assistant          0.5134

  PhD:thesis :: MTech:?            1     done               0.5989  ✓ yes
                                   2     doctors            0.5666
                                   3     healthcare         0.5550

  research:lab :: student:?        1     qp                 0.5106  ~ maybe
                                   2     opt                0.5093
                                   3     autumn             0.5066
```

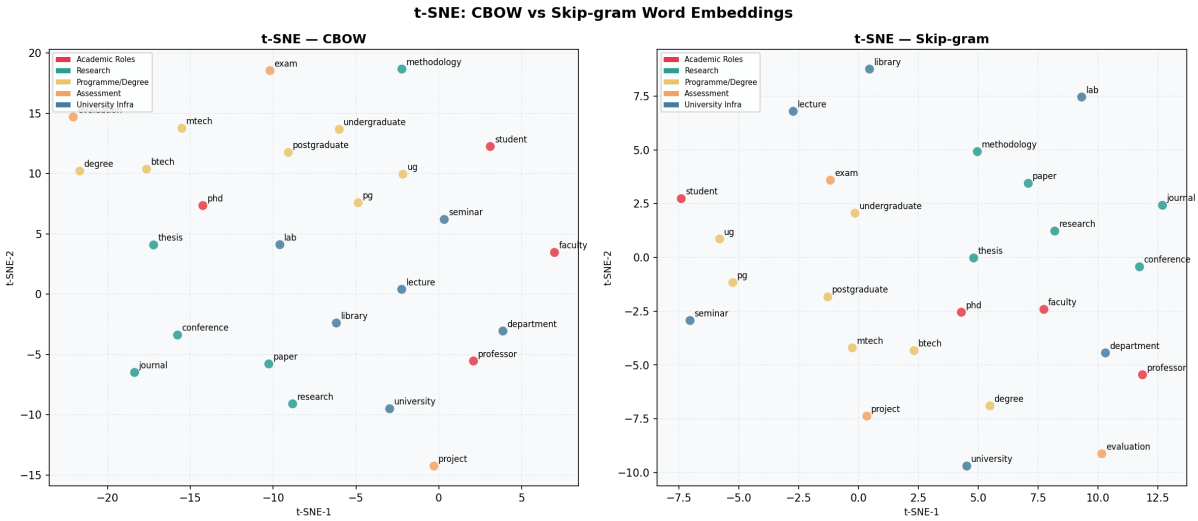
- **Skip-gram:**
  Generates more **diverse but less precise results**. Some outputs are meaningful (*PG* → mtech, professor → master), but many are less relevant (qp, autumn).

➔ Visualization

○ PCA



- ## t-SNE



Problem 2: Character-level name generation using RNN variants

➔ Model information:

Vanilla RNN
Architecture:

- Input sequence is passed through a standard **Recurrent Neural Network (RNN)** layer
- Processes data **sequentially (left → right)**
- The hidden state is updated at each time step
- Final hidden state is passed to a **fully connected (Linear) layer** for output
- Learns patterns based only on **past context**
- **Trainable parameter:** 63,754

Bidirectional LSTM (BLSTM)
Architecture:

- Uses **two LSTM layers**:
  - Forward pass (left → right)
  - Backward pass (right → left)
- Outputs from both directions are **concatenated**
- Combined representation is passed to a **Linear layer**
- Captures **both past and future context** for better sequence understanding
- **Trainable parameter:** 633,354

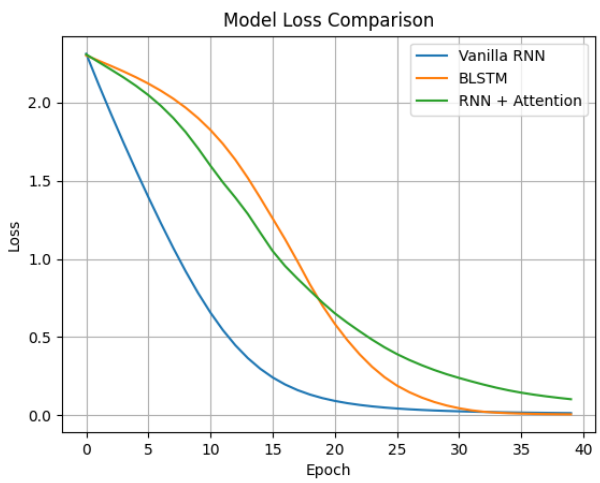
RNN with Attention Mechanism
Architecture:

- Input sequence first processed by a **Vanilla RNN**
- Produces hidden states for all time steps
- An **attention layer** assigns importance weights to each time step
- The weighted sum (context vector) is computed
- Context vector passed to the **Linear layer** for output
- Model learns to **focus on important parts of the sequence**
- **Trainable parameter:** 63,883

RNN: Uses past context only
BLSTM: Uses past + future context
RNN + Attention: Focuses on important tokens dynamically

Hidden Size: 128
Layers: 2
Learning Rate: 0.001

➔ Training loss curve:



➔ Quantitative analysis:

Model	Novelty rate	Diversity
Vanilla RNN	100	0.9300
BLSTM	100	0.9280
RNN_Attention	100	0.9320

➔ Realism of Generated Names

Vanilla RNN
Examples:

- *anmana, raan, nasavi, vitani, larina*

Analysis:

- Produces **short, simple syllable-based names**
- Some names resemble Indian patterns (*nasavi, vitani*)
- But many are **incomplete or unnatural combinations** (*raan, anmana*)

BLSTM
Examples:

- *Raya, Naviri, Sarika, Yamadena, Visaro*

Analysis:

- Better structure due to **bidirectional context**
- More names resemble real Indian names (*sarika, raya*)
- Still contains unrealistic blends (*yamadena, rorakusu*)

RNN + Attention
Examples:

- *kushande, savi, vilana, karadi, nira*

Analysis:

- Slightly more **balanced and structured**
- Produces smoother syllable transitions
- Some names are closer to realistic forms (*savi, nira*)

Github link:

https://github.com/jahanvi0106/Natural_Language_understanding/tree/Assignment2